The critical one-dimensional concave assignment cost is $\boldsymbol{\Theta(\sqrt{n}\log n)}$

A counterexample to the finite $\sqrt{n\log n}$-normalization in arXiv:2305.09234

Automated mathematical discovery run `fa_banach_001`
Lane 9, `agent_lane_09`, model GPT5.6

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Status. Candidate counterexample, likely valid, pending human review. For the uniform law the normalized expectation in the source tends to $+\infty$. Thus this is a negative answer to the intended *finite* limit conjecture. If "limit" is interpreted in the extended real line, the limit instead exists and equals $+\infty$; that wording distinction is preserved throughout.

Abstract

Let $X_1, \dots, X_n, Y_1, \dots, Y_n$ be independent uniform points of $[0,1]$, and let

$$M_n = \min_{\sigma\in S_n} \sum_{i=1}^{n} |X_i - Y_{\sigma(i)}|^{1/2}.$$

Goldman and Trevisan asserted $\limsup \mathbb{E}M_n/\sqrt{n\log n} < \infty$, citing a 2020 transport estimate, and conjectured existence of the associated finite limit. Bobkov and Ledoux's published 2024 correction states that the underlying proposition was false and replaces the critical empirical upper bound by $O(\log n/\sqrt{n})$ for normalized measures. We prove the matching lower bound by a signed multiscale dyadic-tent witness in Kantorovich–Rubinstein duality. Consequently

$$\mathbb{E}M_n \asymp \sqrt{n}\log n, \qquad \frac{\mathbb{E}M_n}{\sqrt{n\log n}} \asymp \sqrt{\log n} \longrightarrow +\infty.$$

The lower bound is elementary and quantitative.

1 Source question and correction

The source is Goldman and Trevisan, *On the concave one-dimensional random assignment problem and Young integration theory*, arXiv:2305.09234 [1]. Item (2) of Section 1.3, split across printed pages 3 and 4, is reproduced below.

(1) For $1/2 < \alpha < 1$, Theorem 1.1 is limited to laws with bounded support. This is because we develop a minimal theory for the Kantorovich-Young problem, valid only in the case of a bounded interval. As with the classical transport problem, to address the case of unbounded intervals, some growth condition should be imposed. In turn, this condition should then be verified in the convergence of the empirical process towards the Brownian bridge, hence extending the results from [23] which we employ in our argument.

(2) A natural question is what happens in the $\alpha = 1/2$ case. It is known [10, Corollary 3.2] that for i.i.d. points $(X_i)_{i=1}^{\infty}$, $(Y_i)_{i=1}^{\infty} \subseteq I$ with common law μ,

$$\limsup_{n\to\infty} \mathbb{E}\left[\mathsf{M}_{1/2}((X_i)_{i=1}^n, (Y_i)_{i=1}^n)\right] / \sqrt{n \log n} < \infty, \tag{1.2}$$

hence a natural conjecture would be that

$$\lim_{n\to\infty} \mathbb{E}\left[\mathsf{M}_{1/2}((X_i)_{i=1}^n, (Y_i)_{i=1}^n)\right] / \sqrt{n \log n}$$

4 M. GOLDMAN AND D. TREVISAN

always exists. In Remark 5.1 below, we prove that if μ is uniform on I, then

$$\liminf_{n\to\infty} \mathbb{E}\left[\mathsf{M}_{1/2}((X_i)_{i=1}^n, (Y_i)_{i=1}^n)\right] / \sqrt{n \log n} > 0. \tag{1.3}$$

(3) Our method is clearly not limited to the assignment problem, as the application to the TSP shows. We conjecture that the connected bipartite κ-factor problem, see [7, 22], should allow for a similar analysis (the case $\kappa = 2$ yields the TSP).

1.4. **Structure of the paper.** In Section 2 we introduce the notation and collect some known facts on Hölder functions, q-variation, optimal transport theory and the convergence of empirical processes towards Brownian bridges. In Section 3, we recall some simple

In transcription, for iid samples with common law μ, the source asserts

$$\limsup_{n\to\infty} \frac{\mathbb{E} M_{1/2}((X_i)_{i=1}^n, (Y_i)_{i=1}^n)}{\sqrt{n \log n}} < \infty, \tag{1}$$

then says that a natural conjecture is that the corresponding limit always exists. It also proves a positive liminf for uniform μ.

The citation supporting (1) was later corrected by its authors. Bobkov and Ledoux state that their Proposition 6.1 "is not correct" and that logarithmic corrections are necessary [2]:

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CORRECTION

Correction: Transport Inequalities on Euclidean Spaces for Non-Euclidean Metrics

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Correction to: Journal of Fourier Analysis and Applications (2020) 26:60
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Proposition 6.1 of the paper is not correct, a fact which impacts a number of statements and conclusions. The note provides the necessary adjustments for the statement and the main corollaries to hold true, including Theorem 1.1. Basically, all the results are still true up to some logarithmic corrections.

The paper under correction is referenced below as [1], and all the notation are taken from it, although a few basic objects and definitions are recalled for convenience. Further results and generalizations of the Fourier analytic bounds obtained in [1] can be stated in terms of Zolotarev distances [3].

1 General Fourier Analytic Bounds

For $d = 1$ and $\alpha = 1/2$, equation (12) of the correction is

$$\mathbb{E}\,\zeta_{1/2}(\mu_n, \nu_n) \le C\,\frac{\log(2n)}{\sqrt{n}}. \tag{2}$$

56 Page 6 of 7 J Fourier Anal Appl (2024) 30:56

When $\alpha \le \frac{1}{2}$, there are additional logarithmic factors. If $\alpha = \frac{1}{2}$, then choosing $T = n$ in (11), we obtain

$$\mathbb{E}\left(\zeta_\alpha(\mu_n, \nu_n)\right) \le c\,\frac{\log(2n)}{\sqrt{n}}. \tag{12}$$

If $\alpha < \frac{1}{2}$, a similar choice leads to

$$\mathbb{E}\left(\zeta_\alpha(\mu_n, \nu_n)\right) \le \frac{c_\alpha}{n^\alpha}\sqrt{\log(2n)}$$

for some constant $c_\alpha > 0$ depending on α only. All these bounds can be sharpened in terms of ψ_2-norms as discussed in [1].

The value $\alpha = \frac{1}{2}$ is therefore critical, in the sense that the rate in the upper bound is changing for smaller values of the parameter α. This threshold phenomenon was already emphasized in [1] in connection with Bernstein's theorem on the absolute convergence of Fourier series for Lipschitz classes Lip(α).

If $d \ge 2$, the rates are different and depend on d. Namely, using (9) we get that, for all $0 < \alpha < 1$,

$$\mathbb{E}\left(\zeta_\alpha(\mu_n, \nu_n)\right) \le \frac{c_\alpha(d)}{n^{\alpha/d}}\sqrt{\log(2n)} \tag{13}$$

with some constants $c_\alpha(d)$ depending on α and d only. This bound should replace the

Since $M_n = n\zeta_{1/2}(\mu_n, \nu_n)$, the corrected bound is $\mathbb{E}M_n \le C\sqrt{n}\log(2n)$. It no longer implies (1).

2 Definitions and result

For probability measures μ, ν on $[0,1]$, write

$$W_d(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \int d(x,y)\, d\pi(x,y), \qquad d(x,y) = |x-y|^{1/2}.$$

The snowflake d is a metric. For a function $f : [0,1] \to \mathbb{R}$, put

$$[f]_{1/2} = \sup_{x\neq y} \frac{|f(x)-f(y)|}{|x-y|^{1/2}}.$$

Kantorovich–Rubinstein duality on the compact metric space $([0,1], d)$ gives [3, Chapter 1]

$$W_d(\mu,\nu) = \sup_{[f]_{1/2}\le 1} \int f\, d(\mu-\nu). \tag{3}$$

Theorem 1 (Sharp critical order and divergent source normalization)**.** *Let $X_1, X_2, \ldots, Y_1, Y_2, \ldots$ be independent* Unif$[0,1]$ *random variables and*

$$M_n = \min_{\sigma\in S_n} \sum_{i=1}^{n} |X_i - Y_{\sigma(i)}|^{1/2}.$$

There are absolute constants $c, C > 0$ such that, for all sufficiently large n,

$$c\sqrt{n}\log n \le \mathbb{E}M_n \le C\sqrt{n}\log n. \tag{4}$$

In particular,

$$\frac{\mathbb{E}M_n}{\sqrt{n\log n}} \longrightarrow +\infty. \tag{5}$$

Thus the uniform law contradicts the finite limsup assertion (1) *and the intended finite-limit conjecture following it. More precisely, that ratio is* $\Theta(\sqrt{\log n})$*.*

3 Proof intuition

The concavity of $t^{1/2}$ makes the optimal permutation difficult to describe directly. The key simplification is that this cost is also a metric. Duality therefore lets us lower-bound the optimal assignment by exhibiting one 1/2-Hölder function.

At every dyadic scale, place a unit tent on each cell. After seeing the two samples, choose each tent's sign to agree with the corresponding empirical discrepancy, and multiply all level-j tents by $2^{-j/2}$. This is the critical coefficient: arbitrary signs and arbitrarily many levels still give a uniformly 1/2-Hölder sum. A tent on a cell of length h sees a discrepancy of expected size at least a constant times $\sqrt{nh}$. There are h^{-1} cells and the coefficient is $\sqrt{h}$, so each level contributes order $\sqrt{n}$. The levels from scale one through scale $1/n$ then add, producing the missing $\log n$.

4 The multiscale lower bound

Let

$$\mu_n = \frac{1}{n}\sum_{i=1}^{n}\delta_{X_i}, \qquad \nu_n = \frac{1}{n}\sum_{i=1}^{n}\delta_{Y_i}.$$

The finite transport linear program has an optimal permutation (equivalently, use the Birkhoff–von Neumann theorem), so (3) yields, pointwise in the samples,

$$M_n = nW_d(\mu_n,\nu_n) = \sup_{[f]_{1/2}\le 1}\sum_{i=1}^{n}(f(X_i)-f(Y_i)). \tag{6}$$

For $j \ge 0$ and $0 \le k < 2^j$, let

$$I_{j,k} = [k2^{-j},(k+1)2^{-j}]$$

and define the unit tent

$$\phi_{j,k}(x) = \max\left\{1 - 2^{j+1}\left|x-(k+\tfrac{1}{2})2^{-j}\right|, 0\right\}.$$

These are the usual dyadic Faber–Schauder tents [4]. At a fixed level their interiors are disjoint and they vanish at the cell endpoints.

Lemma 2 (Uniform signed Hölder bound)**.** *For every integer $J \ge 1$ and every choice $\varepsilon_{j,k} \in \{-1,1\}$,*

$$f_J(x) = \sum_{j=0}^{J-1} 2^{-j/2}\sum_{k=0}^{2^j-1}\varepsilon_{j,k}\phi_{j,k}(x)$$

satisfies $[f_J]_{1/2} < 14$.

Proof. Write the level-j summand as g_j. Disjointness within a level gives $\|g_j\|_\infty \le 2^{-j/2}$, while its continuous piecewise-affine graph has slopes of absolute value at most $2^{j/2+1}$. Hence, for $r = |x-y|$,

$$|g_j(x)-g_j(y)| \le \min\{2\,2^{-j/2},\ 2^{j/2+1}r\}. \tag{7}$$

If $0 < r \le 1$, choose $m \ge 0$ so that $2^{-(m+1)} < r \le 2^{-m}$. Using the slope bound for $j \le m$ and the amplitude bound for $j > m$, and harmlessly extending both finite sums to the corresponding infinite ones, gives

$$\begin{aligned} |f_J(x) - f_J(y)| &\le 2r \sum_{j=0}^{m} 2^{j/2} + 2 \sum_{j=m+1}^{\infty} 2^{-j/2} \\ &< \left(\frac{2\sqrt{2}}{\sqrt{2}-1} + \frac{2}{1-2^{-1/2}} \right) \sqrt{r} < 14\sqrt{r}. \end{aligned}$$

Here $r2^{m/2} \le \sqrt{r}$ and $2^{-(m+1)/2} < \sqrt{r}$. This proves the claim. □

For each cell define the empirical tent discrepancy

$$S_{j,k} = \sum_{i=1}^{n} \phi_{j,k}(X_i) - \sum_{i=1}^{n} \phi_{j,k}(Y_i).$$

Choose $\varepsilon_{j,k} = \operatorname{sgn}(S_{j,k})$, with either sign when $S_{j,k} = 0$. The test function $f_J/14$ is admissible in (6); crucially, dependence on the realized samples is legitimate because the supremum is evaluated separately for every realization. Thus

$$M_n \ge \frac{1}{14} \sum_{j=0}^{J-1} 2^{-j/2} \sum_{k=0}^{2^j-1} |S_{j,k}|. \tag{8}$$

Lemma 3 (One-cell discrepancy)**.** *There is an absolute constant $c_0 > 0$ such that, for any dyadic cell of length h and its unit tent ϕ,*

$$\mathbb{E}\left|\sum_{i=1}^{n} (\phi(X_i) - \phi(Y_i))\right| \ge c_0\sqrt{nh}$$

whenever $nh \ge 12$. One may take $c_0 = (16\sqrt{12})^{-1}$.

Proof. For $U \sim \mathrm{Unif}[0,1]$, direct integration of the triangular function gives

$$\mathbb{E}\phi(U) = \frac{h}{2}, \qquad \mathbb{E}\phi(U)^2 = \frac{h}{3}.$$

For independent U, V, put $Z = \phi(U) - \phi(V)$. Then Z is centered and symmetric, $|Z| \le 1$, and

$$v := \mathbb{E}Z^2 = 2\left(\frac{h}{3} - \frac{h^2}{4}\right) \ge \frac{h}{6}.$$

For $S = \sum_{i=1}^{n} Z_i$, independence and centering give

$$\mathbb{E}S^4 = n\mathbb{E}Z^4 + 3n(n-1)v^2 \le nv + 3n^2v^2.$$

If $nh \ge 12$, then $nv \ge 2$, and therefore $\mathbb{E}S^4 \le 4(nv)^2$. Applying Paley–Zygmund to S^2 at threshold $\frac{1}{2}\mathbb{E}S^2$ gives

$$\Pr\{S^2 \ge \tfrac{1}{2}nv\} \ge \frac{1}{16}.$$

Consequently

$$\mathbb{E}|S| \ge \frac{1}{16}\sqrt{\frac{nv}{2}} \ge \frac{1}{16\sqrt{12}}\sqrt{nh}.$$

□

Proof of Theorem 1. For $n \geq 12$, let

$$L_n = 1 + \lfloor \log_2(n/12) \rfloor$$

and use (8) with $J = L_n$. At every included level j, the cell length $h = 2^{-j}$ satisfies $nh \geq 12$. Lemma 3 and the 2^j cells yield

$$\begin{aligned} 2^{-j/2} \sum_{k=0}^{2^j-1} \mathbb{E}|S_{j,k}| &\geq 2^{-j/2}\, 2^j c_0 \sqrt{n 2^{-j}} \\ &= c_0 \sqrt{n}. \end{aligned}$$

Taking expectations in (8) and summing levels gives

$$\mathbb{E} M_n \geq \frac{c_0}{14} L_n \sqrt{n} \geq c\sqrt{n} \log n$$

for all sufficiently large n. This proves the lower bound without any external asymptotic theorem.

For the upper bound, apply Bobkov–Ledoux's corrected equation (12), (2), to the two empirical measures. Since $M_n = n\zeta_{1/2}(\mu_n, \nu_n)$,

$$\mathbb{E} M_n \leq C\sqrt{n} \log(2n),$$

which is $O(\sqrt{n} \log n)$. This proves (4). Dividing by $\sqrt{n \log n}$ proves (5) and the stated $\Theta(\sqrt{\log n})$ refinement. □

5 Verification and audit

The reusable checker `code/verify_multiscale_witness.py` performs three supplementary tests:

1. it checks the exact moment inequalities at fourteen dyadic levels;
2. for twelve random sign systems on ten levels, it samples three million point pairs on a 4097-point grid and checks the Hölder ratio against 14;
3. for twenty random instances at each of $n = 32, 64, 128, 256$, it compares the normalized witness against the exact Hungarian assignment cost.

The command and complete output are recorded in `verification.md`. These finite checks are regression evidence only; Lemmas 2 and 3 are the proof.

The following proof-sensitive points were audited explicitly.

- The cost $|x - y|^{1/2}$ is a metric, so the standard Kantorovich–Rubinstein duality applies; this would fail for an arbitrary concave cost.
- The witness signs may depend on the samples because duality is a pointwise supremum before expectation.
- Adjacent tents with opposite signs cause no discontinuity: every tent vanishes at its cell endpoints, and the global slope bound used in (7) remains valid.
- No independence between different dyadic cells or levels is used. Linearity of expectation is enough after the one-cell estimate.
- The supporting upper bound is cited from the 2024 correction, not from the invalid 2020 formulation used in the source paper.

6 Scope, novelty search, and limitations

The theorem settles the critical order for the uniform law and therefore supplies a counterexample to the asserted finite limsup and intended finite limit. It does not classify all common laws μ, nor does it claim that the normalized sequence lacks an extended-real limit. For this example it has the extended-real limit $+\infty$.

A bounded novelty search was conducted through 13 August 2026 using the four run indexes, arXiv/web exact-title searches, citation searches around arXiv:2305.09234 and the 2024 correction, and combinations of the phrases "$M_{1/2}$", "$\sqrt{n}\log n$", "Zolotarev empirical measures", and "Faber–Schauder matching". The search found the source, the correction, and unrelated matching work, but no paper explicitly connecting the correction to the source conjecture or proving the dyadic-tent lower bound. This is evidence of novelty, not a guarantee. A human should repeat the literature search, check the Hölder constant and fourth-moment step, and decide whether the source's wording warrants describing the result as a "counterexample" or more narrowly as a "sharp critical-rate correction".

References

[1] M. Goldman and D. Trevisan, *On the concave one-dimensional random assignment problem and Young integration theory*, arXiv:2305.09234 (2023).

[2] S. G. Bobkov and M. Ledoux, *Correction: Transport inequalities on Euclidean spaces for non-Euclidean metrics*, Journal of Fourier Analysis and Applications **30** (2024), article 56, doi:10.1007/s00041-024-10105-y.

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[4] J. Schauder, *Zur Theorie stetiger Abbildungen in Funktionalräumen*, Mathematische Zeitschrift **26** (1927), 47–65.